CLUTCH MASTER CYLINDER REMOVAL/INSTALLATION [F66M-R]

id0510q2156800

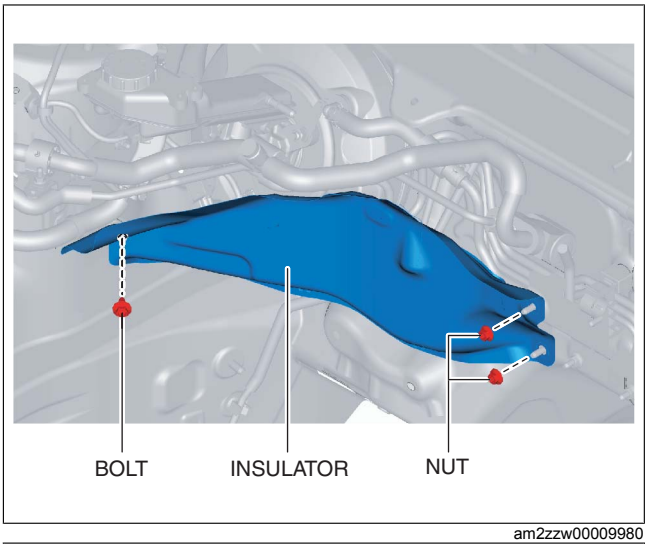
Caution

- **Secure the steering wheel using tape or a cable to prevent the steering shaft from rotating after disconnecting the steering shaft. If the steering wheel rotates after the steering shaft and the steering gear and linkage are disconnected, the internal parts of the clock spring could be damaged.**
- **Do not allow clutch fluid to get on a painted surface. Clutch fluid contains properties which can dissolve the paint. If clutch fluid gets on a painted surface, rinse with water immediately, then wipe the area dry.**

SKYACTIV-G 1.5

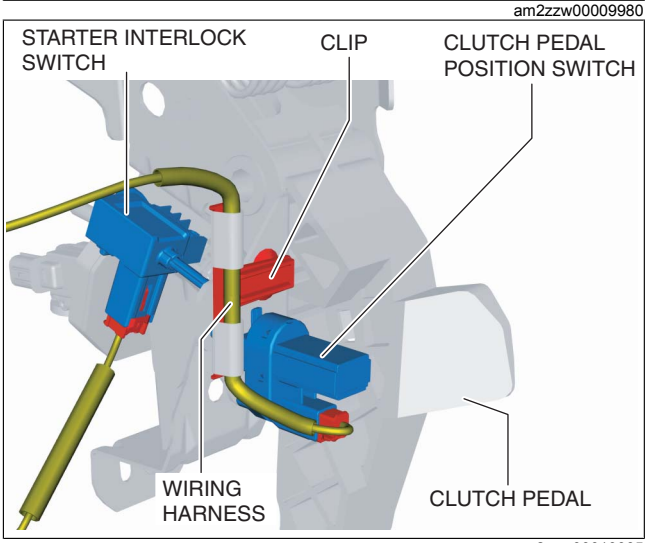
Removal

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)
2. Remove the air cleaner, air hose and fresh air duct as a single unit. (L.H.D.) (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
3. Remove the battery tray and PCM component. (L.H.D.) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
4. Remove the insulator. (R.H.D.)
5. Disconnect the clutch stroke sensor connector. (with i-stop)



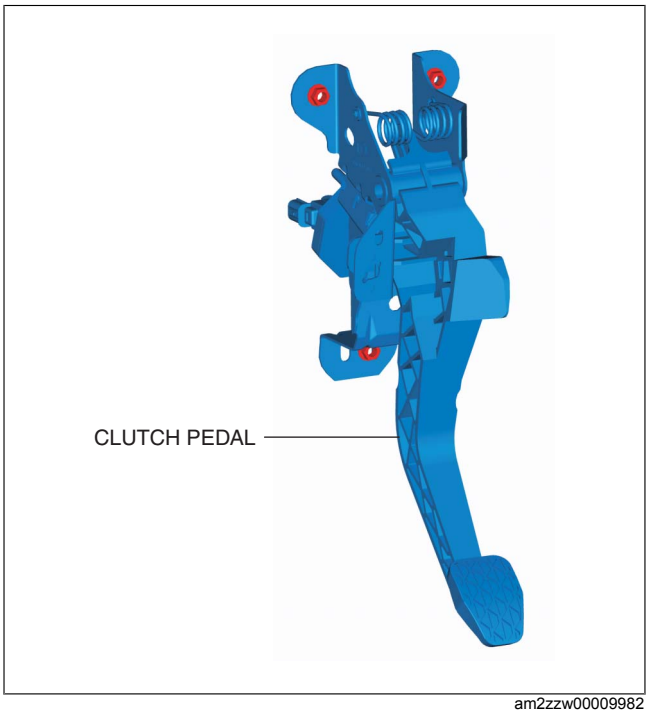
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6. Disconnect the wiring harness from the clutch pedal, and set the wiring harness aside. (L.H.D.)
7. Remove the clutch pedal position switch. (See CLUTCH PEDAL POSITION SWITCH REMOVAL/INSTALLATION [F66M-R].)
8. Remove the starter interlock switch. (See STARTER INTERLOCK SWITCH REMOVAL/INSTALLATION [F66M-R].)
9. Remove the joint cover. (See STEERING WHEEL AND COLUMN REMOVAL/INSTALLATION.)
10. Disconnect the intermediate shaft from the steering gear and linkage. (See STEERING WHEEL AND COLUMN REMOVAL/INSTALLATION.)
11. Disconnect the clutch pipe No.1 from clutch master cylinder, and plug it to avoid clutch fluid leakage. (See CLUTCH PIPE AND HOSE REMOVAL/INSTALLATION [F66M-R].)
12. Disconnect the clutch reserve hose from clutch master cylinder, and plug it to avoid clutch fluid leakage. (See CLUTCH PIPE AND HOSE REMOVAL/INSTALLATION [F66M-R].)
13. Remove the clutch master cylinder using the following procedure:
    (1) Remove the clutch pedal and the clutch master cylinder as a single unit.



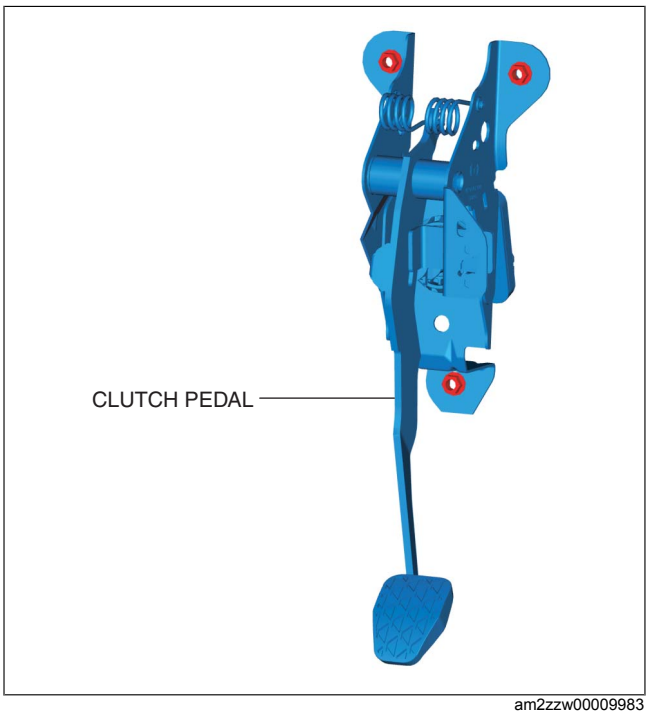
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L.H.D.



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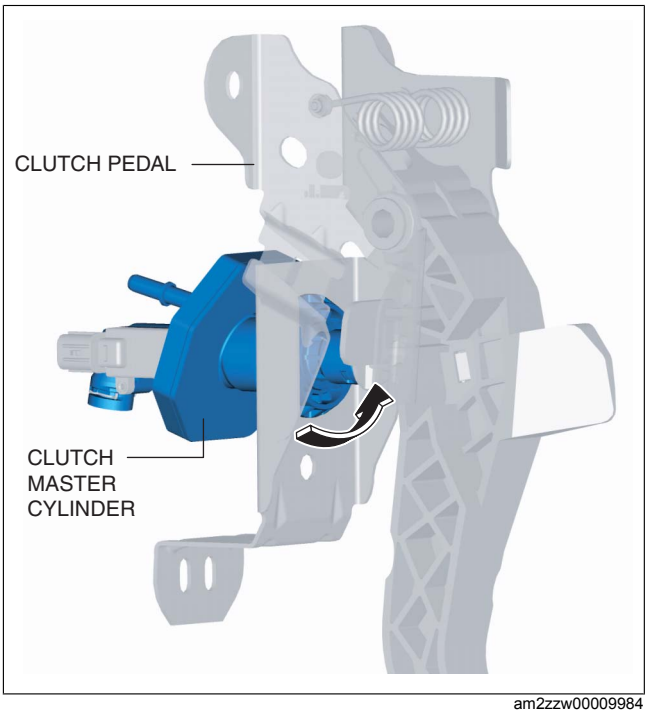
R.H.D.



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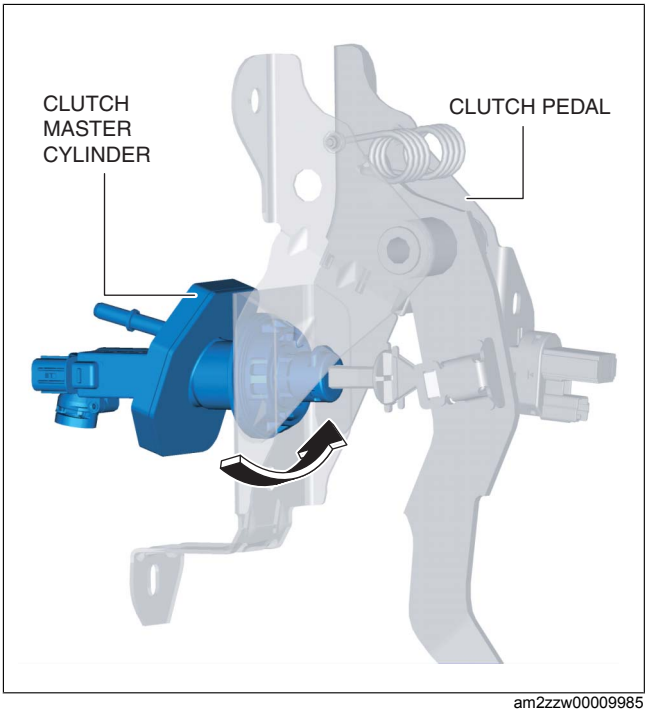
(2) Rotate the clutch master cylinder in the direction of the arrow shown in the figure to disconnect the clutch master cylinder (body) and clutch pedal.

L.H.D.



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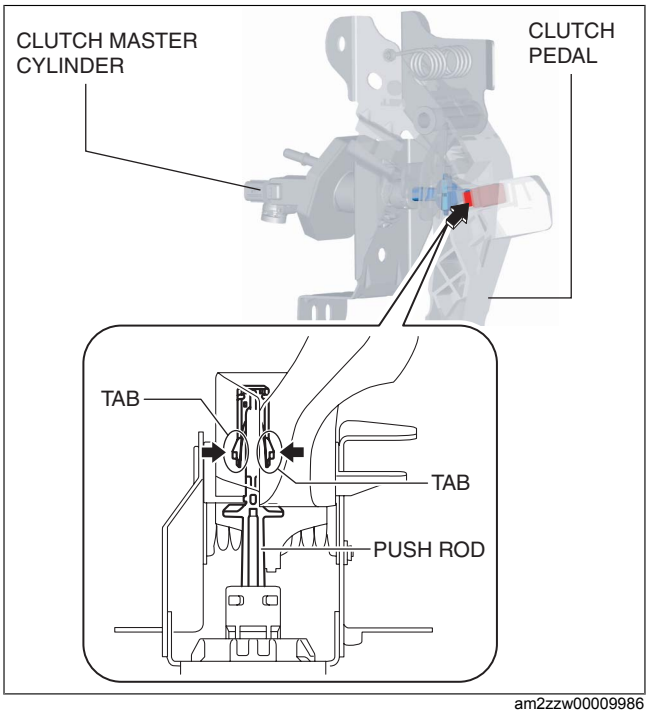
R.H.D.



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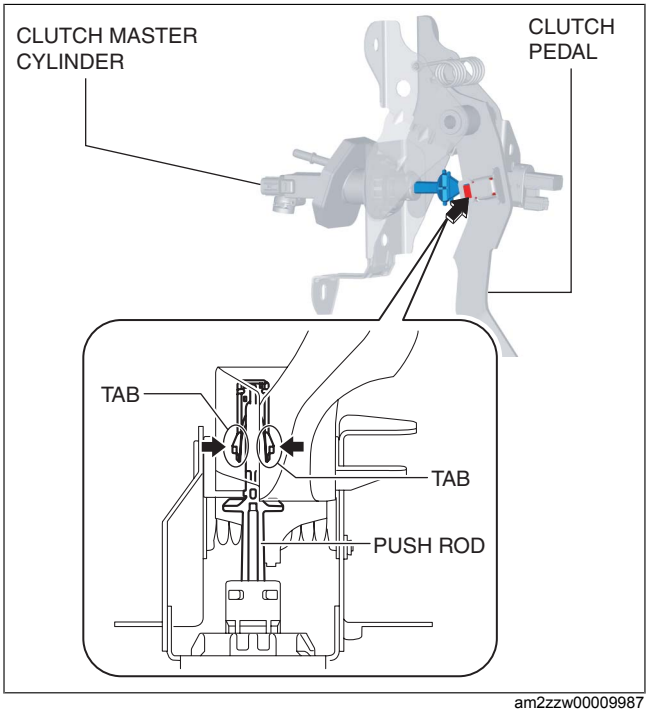
(3) Disengage the tabs of the push rod and remove the clutch master cylinder.

L.H.D.



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R.H.D.

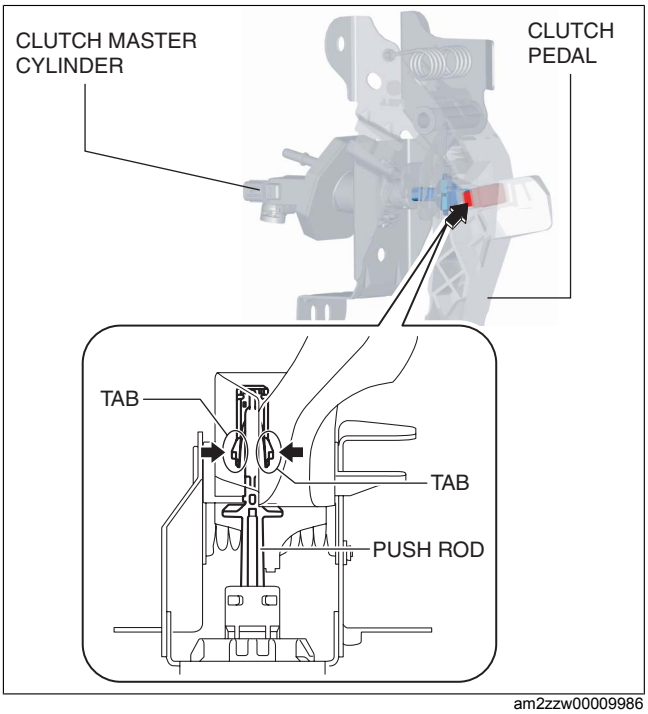


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Installation

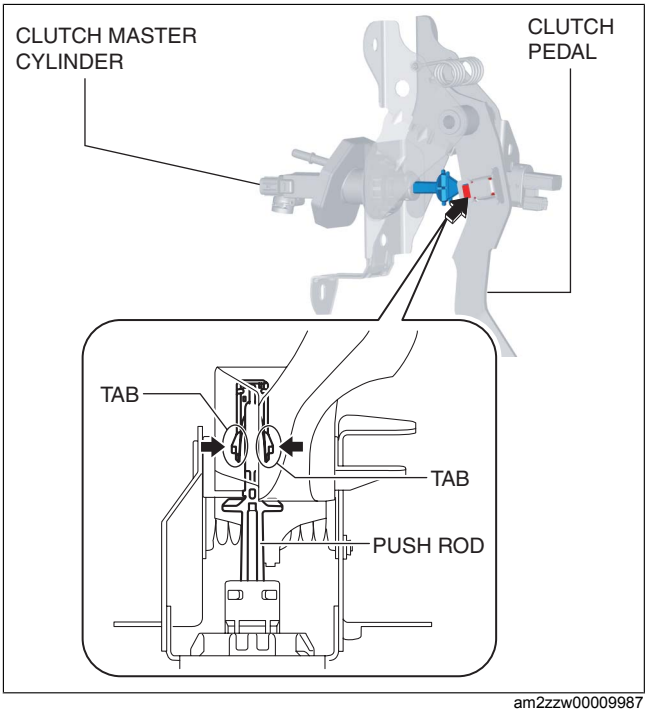
1. Install the clutch master cylinder using the following procedure:
   (1) Engage the tabs of the push rod with the clutch pedal as shown in the figure.

L.H.D.



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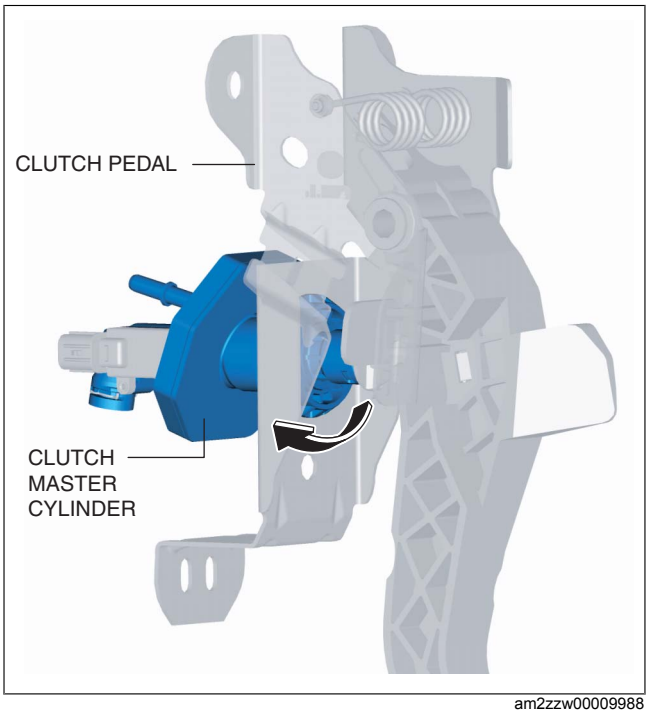
R.H.D.



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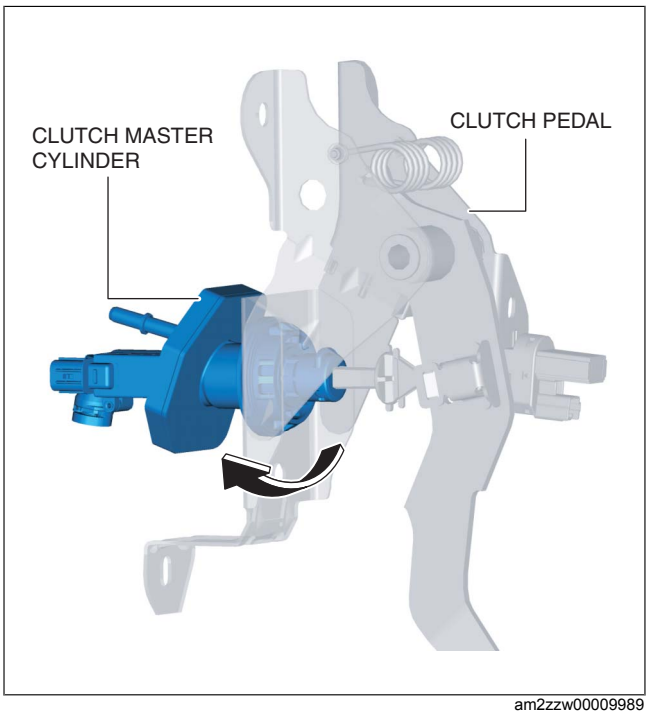
(2) Rotate the clutch master cylinder in the direction shown until it stops.

L.H.D.



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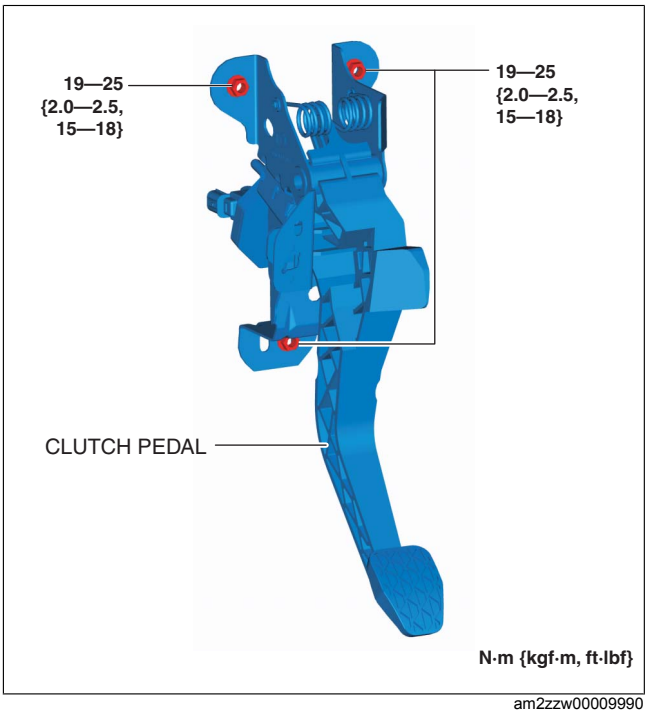
R.H.D.



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(3) Install the clutch pedal and the clutch master cylinder as a single unit.

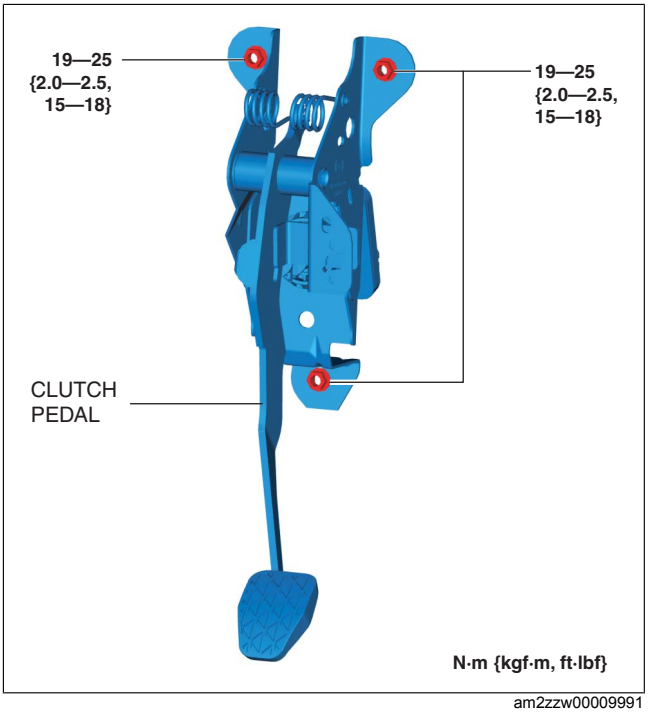
L.H.D.



am2zzw00009990

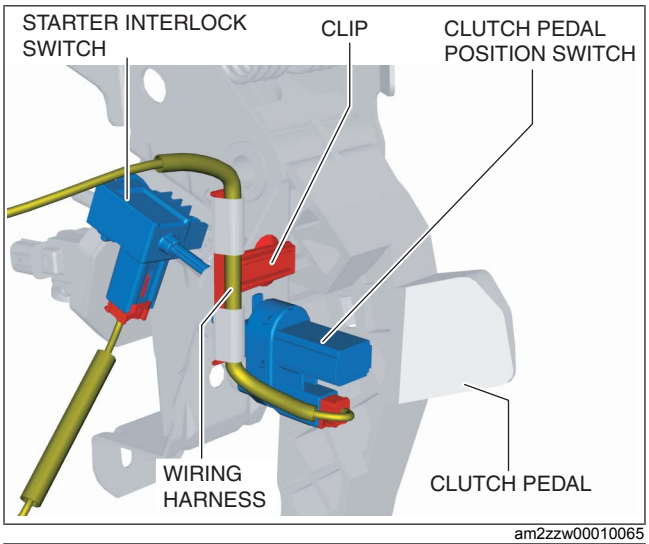
R.H.D.

2. Connect the clutch reserve hose to clutch master cylinder. (See CLUTCH PIPE AND HOSE REMOVAL/INSTALLATION [F66M-R].)
3. Connect the clutch pipe No.1 to clutch master cylinder. (See CLUTCH PIPE AND HOSE REMOVAL/INSTALLATION [F66M-R].)
4. Install the intermediate shaft to the steering gear and linkage. (See STEERING WHEEL AND COLUMN REMOVAL/INSTALLATION.)
5. Install the joint cover. (See STEERING WHEEL AND COLUMN REMOVAL/INSTALLATION.)
6. Bleed the air from the clutch system. (See CLUTCH FLUID REPLACEMENT/AIR BLEEDING [F66M-R].)
7. Inspect the clutch pedal height, and verify that the clutch pedal is installed correctly. (See CLUTCH PEDAL INSPECTION [F66M-R].)
8. Install a new starter interlock switch. (See STARTER INTERLOCK SWITCH REMOVAL/INSTALLATION [F66M-R].)
9. Install a new clutch pedal position switch. (See CLUTCH PEDAL POSITION SWITCH REMOVAL/INSTALLATION [F66M-R].)



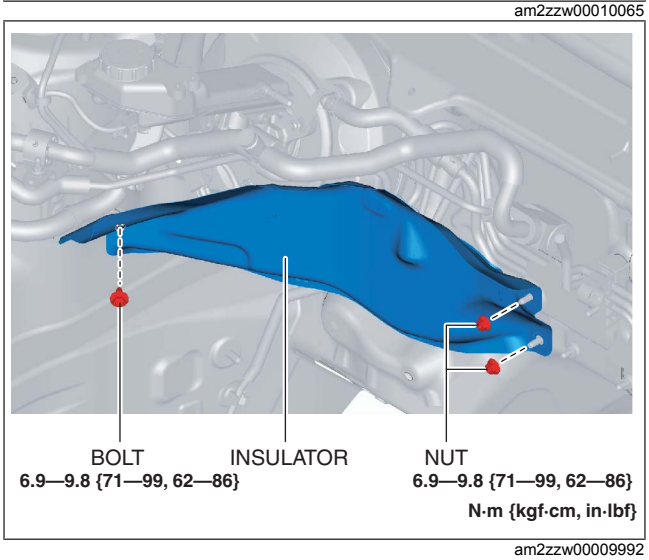
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10. Connect the wiring harness to the clutch pedal. (L.H.D.)
11. Connect the clutch stroke sensor connector. (with i-stop)



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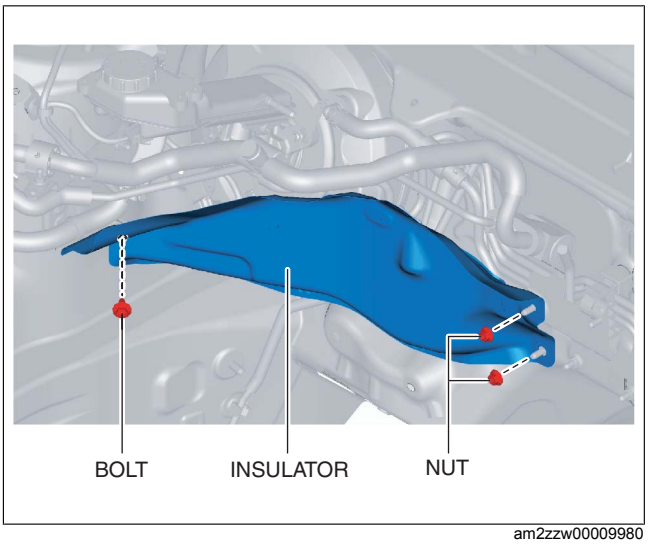
12. Install the insulator. (R.H.D.)
13. Install the air cleaner, air hose and fresh air duct as a single unit. (L.H.D.) (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
14. Install the battery tray and PCM component. (L.H.D.) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
15. Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)
16. Fully depress the clutch pedal, and verify that the engine starts.



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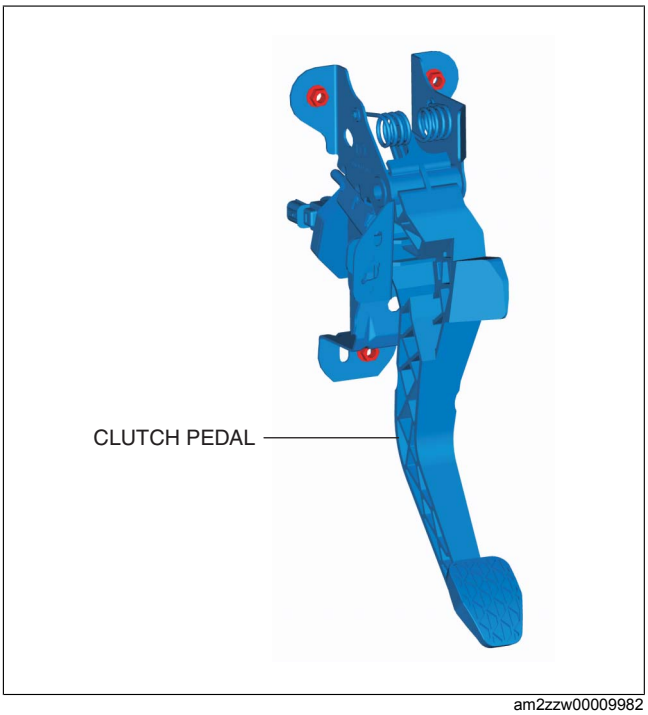
SKYACTIV-D 1.5
Removal
1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)
2. Remove the air cleaner, air hose and fresh air duct as a single unit. (L.H.D.) (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
3. Remove the battery and battery tray. (L.H.D.) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
4. Remove the DC-DC converter (i-ELOOP) with the cables still connected and set it out of the way. (L.H.D.) (with i-ELOOP) (See DC-DC CONVERTER (i-ELOOP) REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].)
5. Remove the windshield wiper arm and blade. (R.H.D.) (See WINDSHIELD WIPER ARM AND BLADE REMOVAL/INSTALLATION.)
6. Remove the cowl grille. (R.H.D.) (See COWL GRILLE REMOVAL/INSTALLATION.)
7. Remove the windshield wiper motor and link. (R.H.D.) (See WINDSHIELD WIPER MOTOR AND LINK REMOVAL/INSTALLATION.)
8. Remove the keyless beeper (with advanced keyless entry system). (R.H.D.) (See KEYLESS BEEPER REMOVAL/INSTALLATION.)
9. Remove the cowl panel. (R.H.D.) (See COWL PANEL REMOVAL/INSTALLATION.)

10. Remove the insulator. (R.H.D.)
11. Disconnect the clutch stroke sensor connector.
12. Remove the clutch pedal position switch. (See CLUTCH PEDAL POSITION SWITCH REMOVAL/INSTALLATION [F66M-R].)
13. Remove the starter interlock switch. (See STARTER INTERLOCK SWITCH REMOVAL/INSTALLATION [F66M-R].)
14. Remove the joint cover. (See STEERING WHEEL AND COLUMN REMOVAL/INSTALLATION.)
15. Disconnect the intermediate shaft from the steering gear and linkage. (See STEERING WHEEL AND COLUMN REMOVAL/INSTALLATION.)
16. Disconnect the clutch pipe No.1 from clutch master cylinder, and plug it to avoid clutch fluid leakage. (See CLUTCH PIPE AND HOSE REMOVAL/INSTALLATION [F66M-R].)
17. Disconnect the clutch reserve hose from clutch master cylinder, and plug it to avoid clutch fluid leakage. (See CLUTCH PIPE AND HOSE REMOVAL/INSTALLATION [F66M-R].)
18. Remove the clutch master cylinder using the following procedure:
    (1) Remove the clutch pedal and the clutch master cylinder as a single unit.



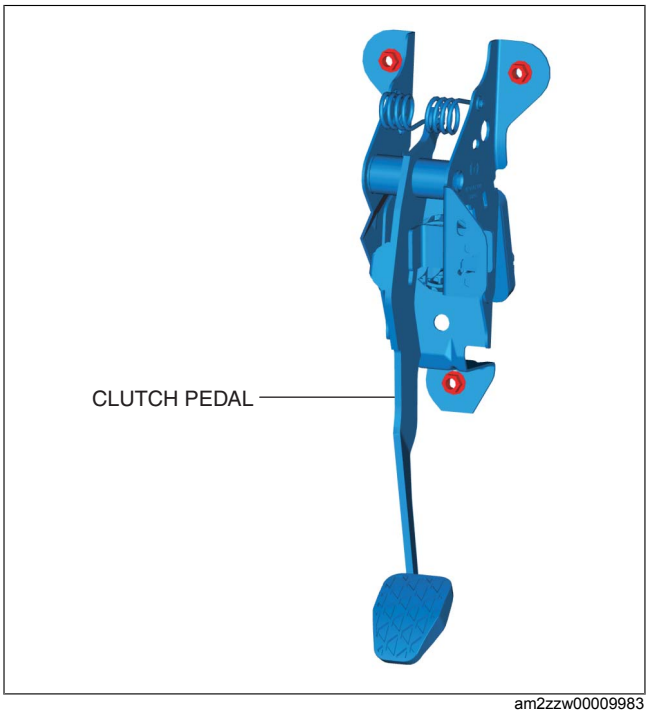
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L.H.D.



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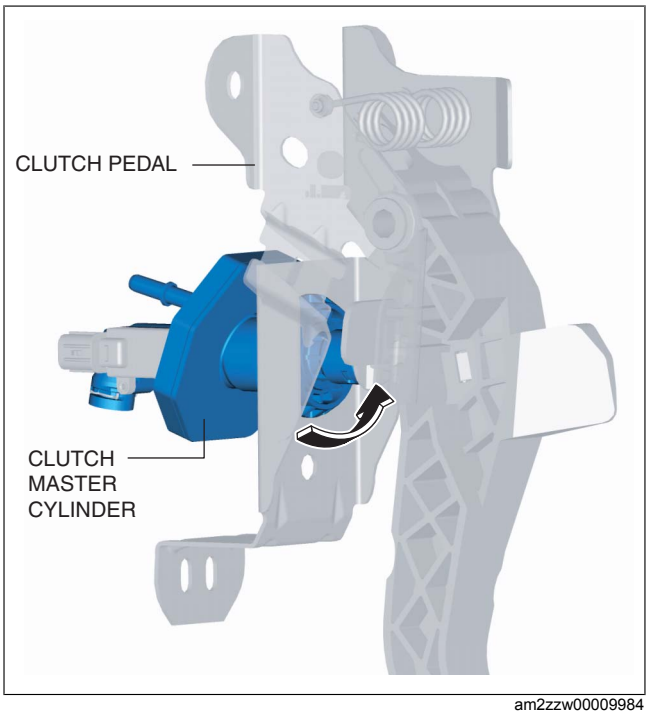
R.H.D.



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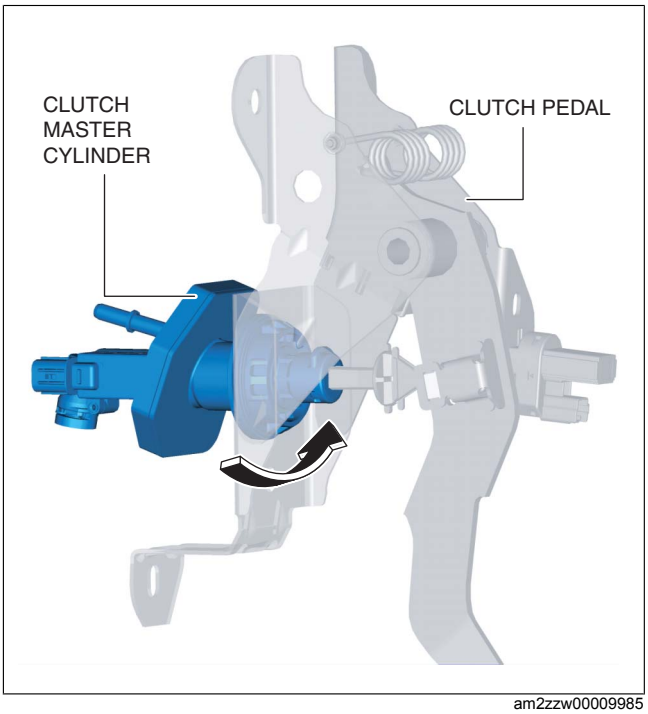
(2) Rotate the clutch master cylinder in the direction of the arrow shown in the figure to disconnect the clutch master cylinder (body) and clutch pedal.

L.H.D.



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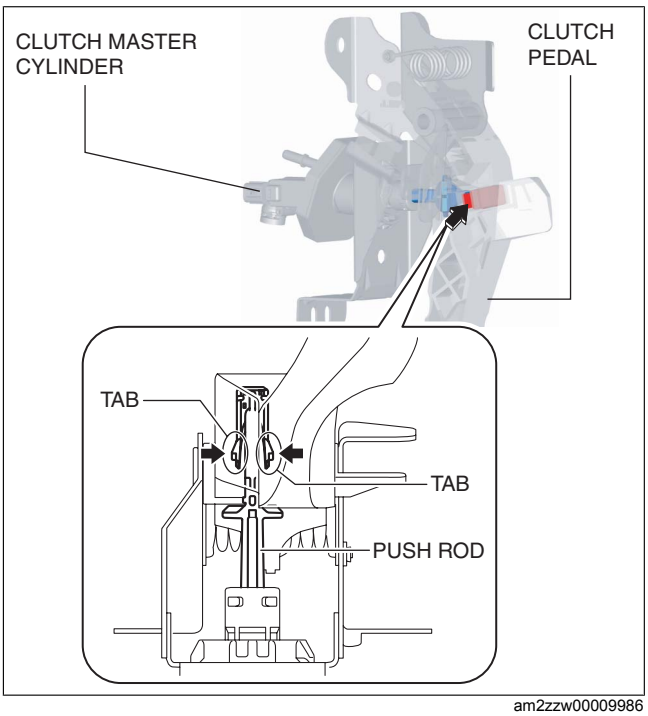
R.H.D.



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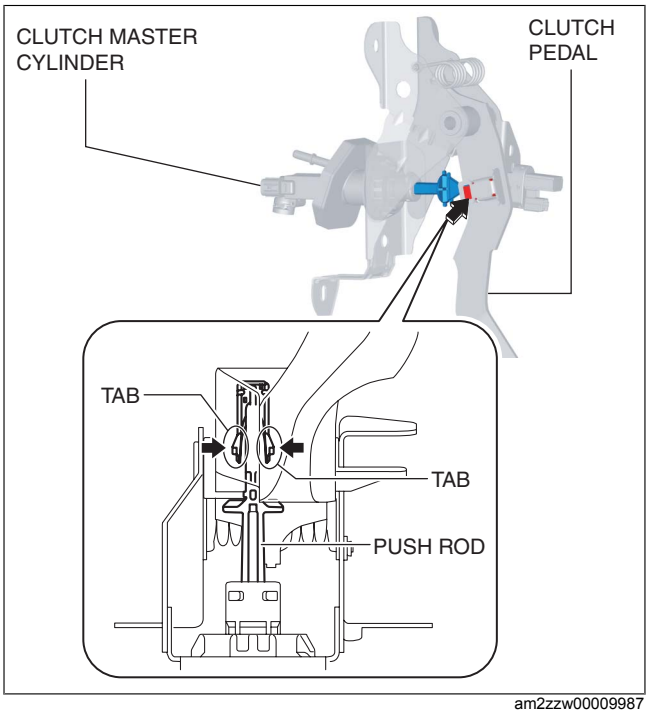
(3) Disengage the tabs of the push rod and remove the clutch master cylinder.

L.H.D.



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R.H.D.

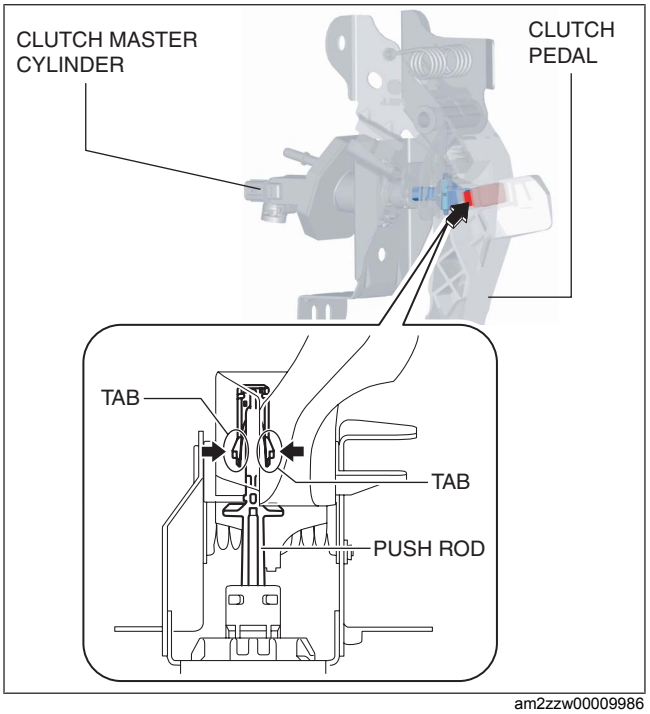


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Installation

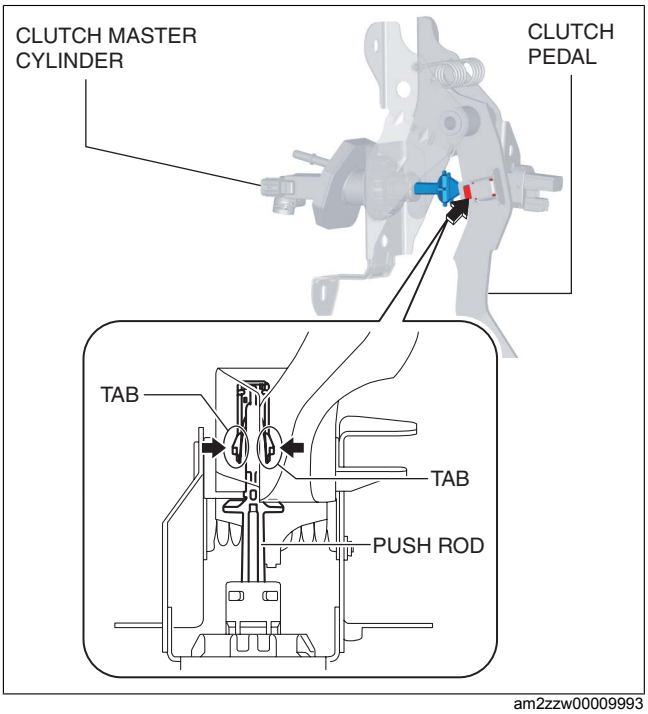
1. Install the clutch master cylinder using the following procedure:
   (1) Engage the tabs of the push rod with the clutch pedal as shown in the figure.

L.H.D.



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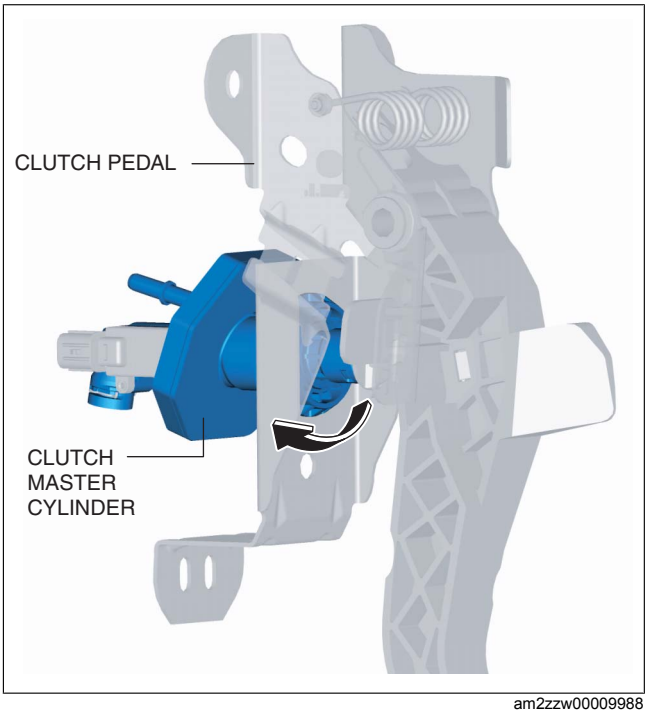
R.H.D.



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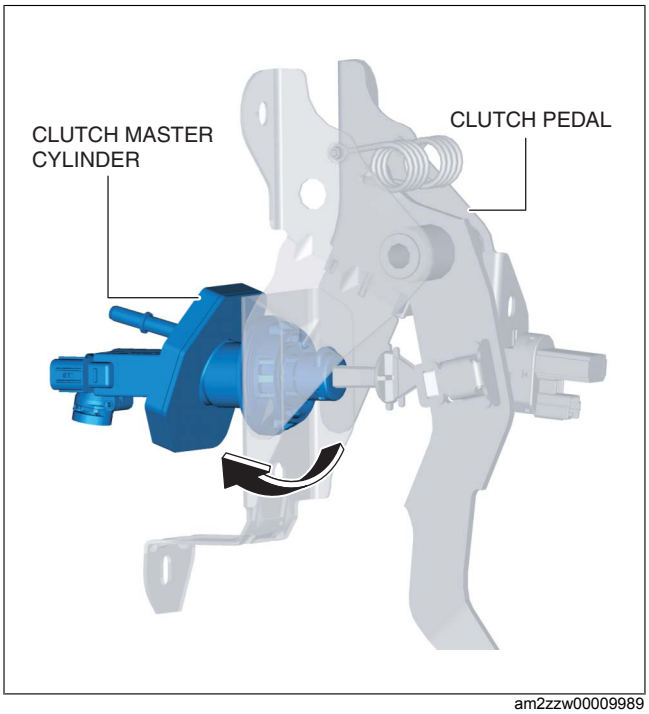
(2) Rotate the clutch master cylinder in the direction shown until it stops.

L.H.D.



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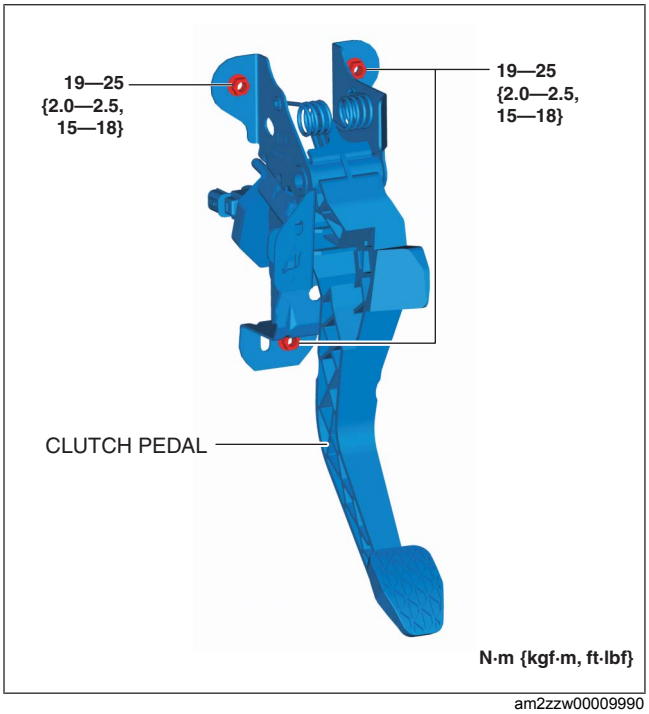
R.H.D.



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(3) Install the clutch pedal and the clutch master cylinder as a single unit.

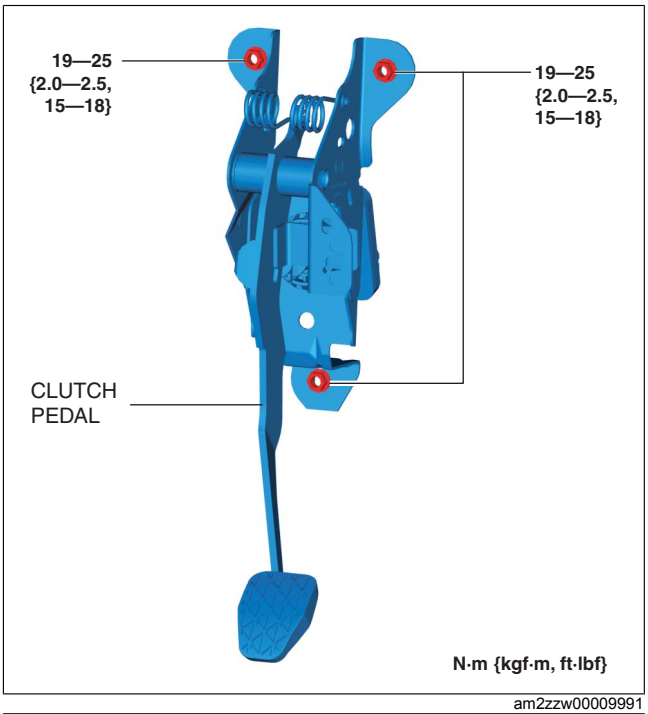
L.H.D.



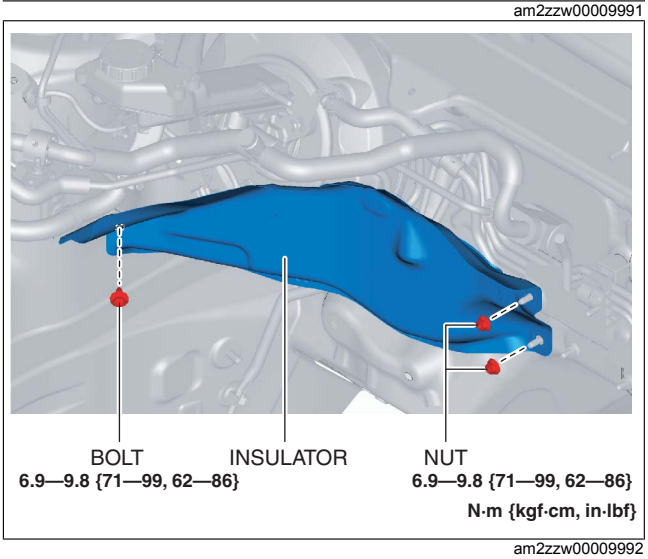
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R.H.D.

2. Connect the clutch reserve hose to clutch master cylinder. (See CLUTCH PIPE AND HOSE REMOVAL/INSTALLATION [F66M-R].)
3. Connect the clutch pipe No.1 to clutch master cylinder. (See CLUTCH PIPE AND HOSE REMOVAL/INSTALLATION [F66M-R].)
4. Install the intermediate shaft to the steering gear and linkage. (See STEERING WHEEL AND COLUMN REMOVAL/INSTALLATION.)
5. Install the joint cover. (See STEERING WHEEL AND COLUMN REMOVAL/INSTALLATION.)
6. Bleed the air from the clutch system. (See CLUTCH FLUID REPLACEMENT/AIR BLEEDING [F66M-R].)
7. Inspect the clutch pedal height, and verify that the clutch pedal is installed correctly. (See CLUTCH PEDAL INSPECTION [F66M-R].)
8. Install a new starter interlock switch. (See STARTER INTERLOCK SWITCH REMOVAL/INSTALLATION [F66M-R].)
9. Install a new clutch pedal position switch. (See CLUTCH PEDAL POSITION SWITCH REMOVAL/INSTALLATION [F66M-R].)
10. Connect the clutch stroke sensor connector.
11. Install the insulator. (R.H.D.)
12. Install the cowl panel. (R.H.D.) (See COWL PANEL REMOVAL/INSTALLATION.)
13. Install the keyless beeper (with advanced keyless entry system). (R.H.D.) (See KEYLESS BEEPER REMOVAL/INSTALLATION.)
14. Install the windshield wiper motor and link. (R.H.D.) (See WINDSHIELD WIPER MOTOR AND LINK REMOVAL/INSTALLATION.)
15. Install the cowl grille. (See COWL GRILLE REMOVAL/INSTALLATION.)
16. Install the windshield wiper arms and blades. (R.H.D.) (See WINDSHIELD WIPER ARM AND BLADE REMOVAL/INSTALLATION.)
17. Install the DC-DC converter (i-ELOOP). (L.H.D.) (with i-ELOOP) (See DC-DC CONVERTER (i-ELOOP) REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].)
18. Install the battery and battery tray. (L.H.D.) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
19. Install the air cleaner, air hose and fresh air duct as a single unit. (L.H.D.) (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
20. Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)
21. Fully depress the clutch pedal, and verify that the engine starts.



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